

MODULE 2 Networking Fundamentals

Phase 1: Networking & Security Foundations • SSC Cybersecurity Training

Phase	Duration
Phase 1	Week 2

Module Overview

This module builds upon the cybersecurity foundations introduced in Module 1 by developing a thorough understanding of computer networking the backbone of all modern digital communication and the primary attack surface that security professionals must defend. Students will learn how data travels across networks, how protocols govern communication, and how network architecture creates security opportunities and vulnerabilities.

By the end of this module, students will have the technical vocabulary and conceptual framework needed to analyze network traffic, understand attacker techniques, and begin thinking like network defenders. This knowledge directly enables the security monitoring, threat detection, and incident response skills covered in later phases.

Learning Objectives

By the end of this module, students will be able to:

- Define networking concepts including IP addressing, subnetting, and routing
- Describe the TCP/IP and OSI models and identify the function of each layer
- Explain the purpose and behavior of common network protocols (DNS, DHCP, HTTP/S, FTP, SSH)
- Differentiate between IPv4 and IPv6 and explain CIDR notation for subnetting
- Identify common network devices and their roles: routers, switches, firewalls, and proxies
- Describe the mechanics of the TCP three-way handshake and stateful connections
- Recognize common network-based attack vectors and how they exploit protocol weaknesses
- Use basic network utilities (ping, traceroute, nslookup) to diagnose connectivity

Key Concepts & Terminology

What is a Network?

A computer network is a collection of interconnected devices that share resources and exchange data. Networks range in scale from two computers connected in a home to billions of devices forming the global Internet. Security professionals must understand networks because virtually every cyberattack involves traversing or manipulating network communication in some way.

Network Types:

Network Type	Description & Security Relevance
LAN (Local Area Network)	A network confined to a single physical location (office, building). Internal attackers and malware spread laterally within LANs. Segmentation is a key defensive technique.
WAN (Wide Area Network)	Networks spanning large geographic areas, including the Internet. WAN links expose organizations to external threats and require encryption for sensitive traffic.
MAN (Metropolitan Area Network)	Larger than a LAN, covers a city or campus. Used by universities, governments, and enterprises. Requires strict access controls across extended infrastructure.
WLAN (Wireless LAN)	Wi-Fi-based networking. Vulnerable to eavesdropping, rogue access points, and Evil Twin attacks. Requires WPA3 encryption and strong authentication.
VPN (Virtual Private Network)	An encrypted tunnel over a public network that creates a secure private channel. Critical for remote workers and protecting sensitive communications.

IP Addressing

Every device on a network is identified by an IP (Internet Protocol) address — a numeric label that enables routing of data packets to the correct destination. Understanding IP addressing is fundamental to network security analysis.

IPv4 Addressing:

IPv4 uses 32-bit addresses expressed as four octets separated by periods (e.g., 192.168.1.100). With only ~4.3 billion possible addresses, IPv4 relies heavily on techniques like NAT (Network Address Translation) and private address ranges.

Address Range	Class	Typical Use
10.0.0.0 – 10.255.255.255	Class A Private	Large enterprise internal networks
172.16.0.0 – 172.31.255.255	Class B Private	Medium enterprise internal networks

192.168.0.0 – 192.168.255.255	Class C Private	Home and small office networks
127.0.0.0 – 127.255.255.255	Loopback	Local host testing (localhost)
0.0.0.0 – 255.255.255.255	Public / Other	Internet-routable addresses assigned by ISPs

CIDR Notation & Subnetting:

CIDR (Classless Inter-Domain Routing) notation expresses a network address and its subnet mask together. For example, 192.168.1.0/24 means the first 24 bits are the network portion, leaving 8 bits for hosts (up to 254 usable host addresses).

CIDR Notation	Description
/8	255.0.0.0 — 16,777,214 usable hosts (Class A scale)
/16	255.255.0.0 — 65,534 usable hosts (Class B scale)
/24	255.255.255.0 — 254 usable hosts (most common LAN)
/30	255.255.255.252 — 2 usable hosts (point-to-point links)
/32	255.255.255.255 — Single host address

IPv6 Addressing:

IPv6 uses 128-bit addresses expressed as eight groups of four hexadecimal digits (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334), providing approximately 340 undecillion unique addresses. IPv6 eliminates the need for NAT and includes built-in IPsec support. Security teams must monitor IPv6 traffic as attackers sometimes use it to bypass legacy IPv4-only security controls.

The TCP/IP Model

The TCP/IP model is the practical framework that governs how data is transmitted across the Internet. Unlike the OSI model (covered in Module 3), the TCP/IP model has four layers that map directly to real-world protocol implementations.

Layer	Function & Key Protocols
4. Application Layer	Provides network services directly to end-user applications. Protocols: HTTP, HTTPS, FTP, SSH, DNS, DHCP, SMTP, SNMP
3. Transport Layer	Manages end-to-end communication and error recovery. Protocols: TCP (reliable, connection-oriented), UDP (fast, connectionless)
2. Internet Layer	Handles logical addressing and routing of packets between networks. Protocols: IP, ICMP, ARP, OSPF, BGP

1. Network Access Layer

Controls hardware and data link operations. Ethernet, Wi-Fi (802.11), MAC addressing

Security Significance of Each Layer:

- **Most cyberattacks target the application layer — SQL injection, XSS, and phishing exploit application protocols** Application Layer
- **Port scanning, SYN floods, and session hijacking target transport-layer weaknesses** Transport Layer
- **IP spoofing, ICMP tunneling, and routing attacks exploit the Internet layer** Internet Layer
- **ARP spoofing, MAC flooding, and rogue device attacks operate at the network access layer** Network Access Layer

TCP vs. UDP

The two primary transport-layer protocols each serve different purposes and carry distinct security implications.

Characteristic	TCP	UDP
Connection Type	Connection-oriented (three-way handshake)	Connectionless (no handshake)
Reliability	Guaranteed delivery with retransmission	No delivery guarantee — best effort
Speed	Slower due to overhead and acknowledgments	Faster — minimal overhead
Order Guarantee	Packets arrive in sequence	Packets may arrive out of order
Common Uses	HTTP/S, SSH, FTP, email (SMTP)	DNS, VoIP, video streaming, online gaming
Security Concern	SYN floods, session hijacking, connection reset attacks	UDP flood DDoS, DNS amplification attacks

The TCP Three-Way Handshake:

TCP establishes a connection through a three-step process before any data is exchanged. Security professionals must understand this process because it is exploited by SYN flood attacks and forms the basis for firewall stateful inspection.

1

SYN – Client sends a SYN (synchronize) packet to the server, requesting a connection and sharing its initial sequence number

2

SYN-ACK – Server replies with SYN-ACK (synchronize-acknowledge), confirming receipt and sharing its own sequence number

3

ACK – Client sends ACK (acknowledge), completing the handshake — the connection is established and data transfer begins

Security Note: SYN Flood Attack

In a SYN flood attack, an attacker sends thousands of SYN packets but never completes the handshake, leaving the server with thousands of half-open connections that exhaust its resources. This is a classic Denial of Service technique. Modern firewalls use SYN cookies to mitigate this attack by only allocating resources after the full handshake completes.

Essential Network Protocols

Security analysts must understand these protocols deeply — both their intended function and how they can be abused by attackers.

DNS — Domain Name System (Port 53):

DNS translates human-readable domain names (google.com) into IP addresses (142.250.80.46). Often called the "phone book of the Internet," DNS is critical infrastructure — but also a frequent attack target.

- **Injecting false DNS records to redirect users to malicious websites** DNS Spoofing / Cache Poisoning
- **Encoding malicious data within DNS queries to exfiltrate data or establish covert command-and-control channels** DNS Tunneling
- **Modifying DNS settings on a victim's system or router to redirect all DNS traffic** DNS Hijacking

DHCP — Dynamic Host Configuration Protocol (Ports 67/68):

DHCP automatically assigns IP addresses, subnet masks, default gateways, and DNS server addresses to devices joining a network, eliminating the need for manual IP configuration.

- **An attacker sets up an unauthorized DHCP server to distribute malicious network settings (wrong gateway or DNS) to clients, enabling man-in-the-middle attacks**
Rogue DHCP Server
- **Flooding a DHCP server with requests to exhaust its IP address pool, preventing legitimate devices from obtaining addresses** DHCP Starvation

HTTP & HTTPS (Ports 80 / 443):

HTTP (Hypertext Transfer Protocol) governs communication between web browsers and servers. HTTPS adds TLS/SSL encryption to protect data in transit. From a security standpoint:

- **HTTP traffic is transmitted in cleartext and can be intercepted by anyone on the network path** Always prefer HTTPS
- **Attackers use forged TLS certificates to impersonate legitimate sites — training users to verify certificates is essential** Certificate Validation
- **Attackers exploit HTTP methods (GET, POST, PUT, DELETE) to perform injection attacks, CSRF, and unauthorized data modification** HTTP Methods

SSH — Secure Shell (Port 22):

SSH provides encrypted remote command-line access to servers and network devices, replacing the insecure Telnet protocol. Security implications:

- **Automated tools attempt to guess SSH credentials — strong passwords and public-key authentication are essential mitigations** Brute Force Attacks
- **Attackers use SSH to create encrypted tunnels that bypass firewall restrictions and exfiltrate data** SSH Tunneling
- **Best practice is to prohibit direct root login via SSH to limit the impact of credential compromise** Disable Root SSH

FTP & SFTP (Ports 20/21 and 22):

FTP (File Transfer Protocol) transfers files between hosts but transmits data — including credentials — in plaintext. SFTP (SSH File Transfer Protocol) provides the same functionality over an encrypted SSH connection and should always be used instead of plain FTP.

Common Port Reference Table:

Port Number	Protocol / Service
20/21	FTP (File Transfer Protocol) — file transfer, cleartext
22	SSH (Secure Shell) — encrypted remote access and SFTP
23	Telnet — remote access, cleartext (obsolete, avoid)

25	SMTP (Simple Mail Transfer Protocol) — email sending
53	DNS (Domain Name System) — name resolution
67/68	DHCP — dynamic IP address assignment
80	HTTP — unencrypted web traffic
110	POP3 — email retrieval
143	IMAP — email access and synchronization
443	HTTPS — encrypted web traffic (TLS/SSL)
3389	RDP (Remote Desktop Protocol) — Windows remote access
8080	HTTP Alternate — commonly used by web proxies and testing

Network Devices & Architecture

Understanding network infrastructure helps security analysts map attack surfaces and identify where security controls should be placed.

Device	Role & Security Relevance
Hub	Broadcasts all traffic to all connected ports — no traffic isolation. Obsolete in modern networks. A single connected attacker can capture all traffic (security risk).
Switch	Forwards traffic only to the intended destination based on MAC addresses. Provides basic traffic isolation. Can be attacked via MAC flooding to force hub-like behavior.
Router	Routes packets between different networks based on IP addresses. Makes forwarding decisions using routing tables. Misconfigured routers can expose internal networks.
Firewall	Monitors and controls incoming/outgoing traffic based on security rules. The primary network perimeter defense. Types: stateless, stateful, next-generation (NGFW).
Proxy Server	Acts as an intermediary for client requests. Forward proxies protect internal users; reverse proxies protect servers. Enable logging and content filtering.
IDS/IPS	Intrusion Detection/Prevention Systems monitor traffic for malicious patterns. IDS alerts; IPS actively blocks. Critical components of security monitoring infrastructure.
WAF	Web Application Firewall — filters HTTP/S traffic to protect web applications from injection, XSS, and other web attacks.
Load Balancer	Distributes traffic across multiple servers. Improves availability and can mitigate some DDoS attacks.

Network Segmentation & DMZ:

Network segmentation divides a network into smaller zones with controlled communication between them. This limits lateral movement by attackers who breach one segment. A DMZ (Demilitarized Zone) is a network segment that sits between the internal network and the Internet, hosting public-facing services like web servers while protecting the internal network.

- **Contains breaches an attacker who compromises a DMZ web server cannot directly reach internal systems** Benefit 1
- **Enables granular monitoring traffic between segments can be logged and inspected** Benefit 2
- **Reduces attack surface internal systems are not directly reachable from the Internet** Benefit 3

Common Network-Based Attack Vectors

Security professionals must recognize how attackers exploit network protocols and infrastructure to intercept, redirect, or disrupt communications.

Attack	Description & Mechanism
ARP Spoofing	The attacker sends forged ARP replies to associate their MAC address with a legitimate IP, redirecting traffic through their device (enables MitM attacks on a LAN)
DNS Cache Poisoning	Injecting malicious DNS records into a resolver's cache, causing users to be redirected to attacker-controlled servers
Man-in-the-Middle (MitM)	The attacker secretly intercepts and potentially alters communications between two parties who believe they are communicating directly
SYN Flood (DoS)	Sending thousands of TCP SYN packets without completing the handshake, exhausting server connection resources
UDP Flood (DDoS)	Overwhelming a target with high-volume UDP packets, consuming bandwidth and processing resources
Port Scanning	Systematically probing a target's ports to identify open services and potential vulnerabilities (reconnaissance technique)
IP Spoofing	Forging the source IP address in packet headers to impersonate another host or conceal the attacker's true location
VLAN Hopping	Exploiting misconfigured trunk ports to send traffic to a VLAN the attacker should not have access to

Basic Network Diagnostic Utilities

These command-line utilities are essential tools for security analysts performing network reconnaissance, troubleshooting, and basic investigation.

Command	Purpose & Example Usage
ping	Tests connectivity and measures latency to a host using ICMP echo requests. Example: ping 8.8.8.8 or ping google.com
tracert / traceroute	Maps the path packets take to a destination, showing each hop. Useful for identifying routing anomalies. Example: tracert google.com
nslookup	Queries DNS to resolve domain names to IP addresses (and vice versa). Example: nslookup google.com
dig	Advanced DNS querying tool providing detailed DNS record information. Example: dig google.com MX
netstat	Displays active network connections, listening ports, and routing tables. Example: netstat -an
ipconfig / ifconfig	Shows the current IP configuration of network interfaces. Example: ipconfig /all (Windows) or ifconfig (Linux)
arp	Displays and modifies the ARP table (IP-to-MAC mappings). Example: arp -a to view the ARP cache
nmap	Network scanner for discovering hosts, open ports, and services (covered in depth in Module 16). Example: nmap -sV 192.168.1.0/24

Additional Resources & Reading

Recommended Reading

- **The definitive reference on TCP/IP protocols** TCP/IP Illustrated, Volume 1 by W. Richard Stevens
- **Widely used networking textbook covering protocols in depth** Computer Networking: A Top-Down Approach by Kurose & Ross
- **Practical guide to monitoring and analyzing network traffic** The Practice of Network Security Monitoring by Richard Bejtlich

Online Resources

- Wireshark User Guide – https://www.wireshark.org/docs/wsug_html_chunked/
- Cloudflare Learning Center (Networking) – <https://www.cloudflare.com/learning/network-layer/>
- Professor Messer's CompTIA Network+ Notes – <https://www.professormesser.com/network-plus/>
- Cisco Networking Basics – <https://www.cisco.com/c/en/us/training-events/training-certifications/certifications.html>
- Subnet Calculator – <https://www.subnet-calculator.com>

Practice Platforms

- **Free network simulation tool for building and testing virtual network topologies**
Cisco Packet Tracer
- **Advanced network emulator supporting real Cisco/Juniper images for hands-on practice** GNS3
- **Guided online labs covering networking fundamentals in a cybersecurity context**
TryHackMe — Pre-Security Path